🎓 Capstone Project - Deep Learning

Programa de Especialización en Inteligencia Artificial para los Negocios

📋 Información del Proyecto

Campo	Información
Nombre del Estudiante	Celso Luis Valiente Morales
Título del Proyecto	Modelo Predictivo de Tamaño de Empresa
Fecha de Entrega	16/03/2026
Profesor	Carlos Mariño

Este notebook entrena modelos de Machine Learning y Deep Learning para **predecir el tamaño de una empresa**.

1. Importar librerías

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

2. Cargar dataset

```
df = pd.read_csv('dataset_empresas.csv')

# Si el dataset es muy grande, puedes activar esta línea para usar una muestra
# df = df.sample(200000, random_state=42)

df.head()
```

	UBIGEO	TAMANO	RANGO_VENTA	RANGO_TRABAJADOR	X1	Y1	CIIU
0	180101	Microempresa	[0 - 150]	[0 - 5]	-70.9357	-17.19270	5520
1	220701	Microempresa	[0 - 150]	[0 - 5]	-76.3311	-6.92216	5122
2	150117	Microempresa	[0 - 150]	[0 - 5]	-77.0841	-11.99820	5234
3	210101	Microempresa	[0 - 150]	[0 - 5]	-70.0237	-15.83870	6120
4	180101	Microempresa	[0 - 150]	[0 - 5]	-70.9367	-17.19300	5139

3. Preparación de datos

```
df_model = df[['UBIGEO', 'RANGO_VENTA', 'RANGO_TRABAJADOR', 'CIIU', 'TAMANO']].copy()

le_dict = {}
for col in df_model.columns:
    le = LabelEncoder()
    df_model[col] = le.fit_transform(df_model[col].astype(str))
    le_dict[col] = le
```

```
X = df_model.drop('TAMANO', axis=1)
y = df_model['TAMANO']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print('Train shape:', X_train.shape)
print('Test shape:', X_test.shape)
```

```
Train shape: (1346888, 4)
Test shape: (336722, 4)
```

4. Modelo Random Forest

```
rf = RandomForestClassifier(random_state=42, n_estimators=100)
rf.fit(X_train, y_train)

pred_rf = rf.predict(X_test)
accuracy_rf = accuracy_score(y_test, pred_rf)

print('Accuracy Random Forest:', accuracy_rf)
```

```
Accuracy Random Forest: 1.0
```

5. Modelo Gradient Boosting

```
gb = GradientBoostingClassifier(random_state=42)
gb.fit(X_train, y_train)

pred_gb = gb.predict(X_test)
accuracy_gb = accuracy_score(y_test, pred_gb)

print('Accuracy Gradient Boosting:', accuracy_gb)
```

```
Accuracy Gradient Boosting: 1.0
```

6. Modelo Deep Learning

```
model = Sequential()
model.add(Dense(64, activation='relu', input_dim=X_train.shape[1]))
model.add(Dense(32, activation='relu'))
model.add(Dense(16, activation='relu'))
model.add(Dense(4, activation='softmax'))

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    X_train,
    y_train,
    epochs=20,
    batch_size=256,
    validation_split=0.2,
    verbose=1
)
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Epoch 1/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 14s 3ms/step - accuracy: 0.9759 - loss: 0.2043 - val_accuracy: 0.9978 - v
Epoch 2/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 20s 3ms/step - accuracy: 0.9968 - loss: 0.0220 - val_accuracy: 0.9985 - v
Epoch 3/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9979 - loss: 0.0157 - val_accuracy: 0.9989 - v
Epoch 4/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 20s 3ms/step - accuracy: 0.9986 - loss: 0.0097 - val_accuracy: 0.9995 - v
Epoch 5/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9992 - loss: 0.0048 - val_accuracy: 0.9996 - v
Epoch 6/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 14s 3ms/step - accuracy: 0.9989 - loss: 0.0110 - val_accuracy: 0.9997 - v
Epoch 7/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9993 - loss: 0.0043 - val_accuracy: 1.0000 - v
Epoch 8/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9993 - loss: 0.0034 - val_accuracy: 1.0000 - v
Epoch 9/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9992 - loss: 0.0079 - val_accuracy: 0.9994 - v
Epoch 10/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 21s 3ms/step - accuracy: 0.9998 - loss: 0.0015 - val_accuracy: 0.9999 - v
Epoch 11/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9995 - loss: 0.0051 - val_accuracy: 0.9998 - v
Epoch 12/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9999 - loss: 7.3364e-04 - val_accuracy: 1.0000
Epoch 13/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9999 - loss: 8.4268e-04 - val_accuracy: 1.0000
Epoch 14/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9998 - loss: 0.0017 - val_accuracy: 1.0000 - v
Epoch 15/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 15s 3ms/step - accuracy: 0.9998 - loss: 0.0027 - val_accuracy: 1.0000 - v
Epoch 16/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9997 - loss: 0.0049 - val_accuracy: 1.0000 - v
Epoch 17/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9995 - loss: 0.0074 - val_accuracy: 1.0000 - v
Epoch 18/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 21s 3ms/step - accuracy: 0.9997 - loss: 0.0017 - val_accuracy: 1.0000 - v
Epoch 19/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 12s 3ms/step - accuracy: 0.9999 - loss: 0.0015 - val_accuracy: 1.0000 - v
Epoch 20/20
4210/4210 ━━━━━━━━━━━━━━━━━━━━ 13s 3ms/step - accuracy: 0.9998 - loss: 0.0029 - val_accuracy: 1.0000 - v
```

Gráfico de accuracy del entrenamiento

```
plt.figure(figsize=(8,5))
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Accuracy del modelo durante el entrenamiento')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.show()
```

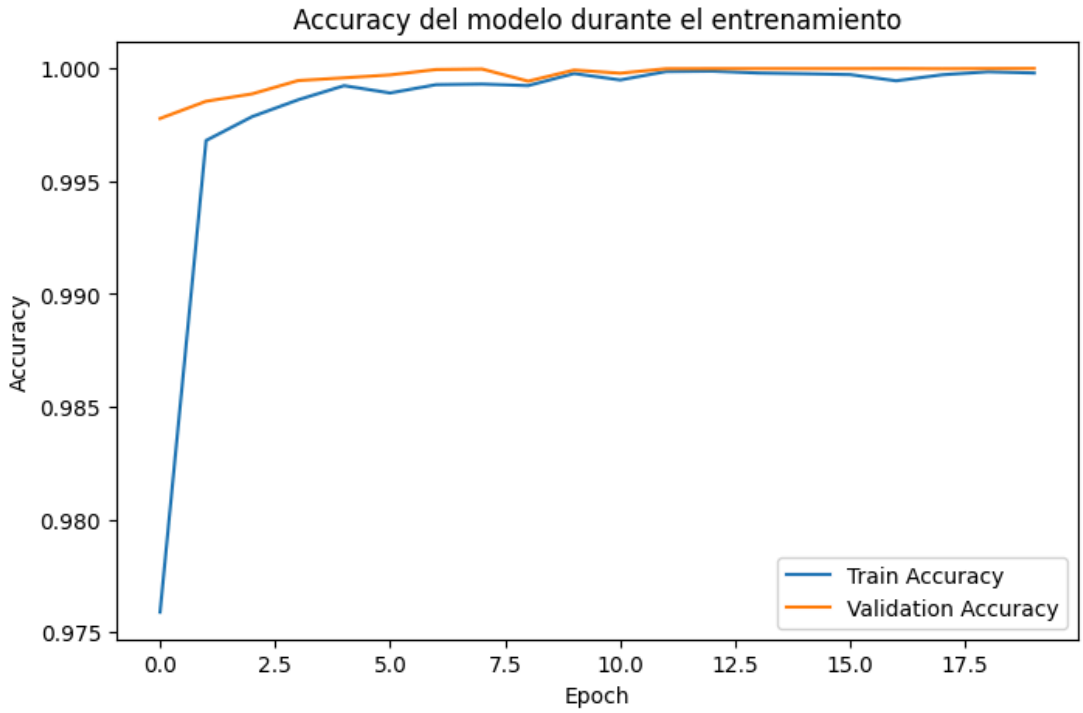
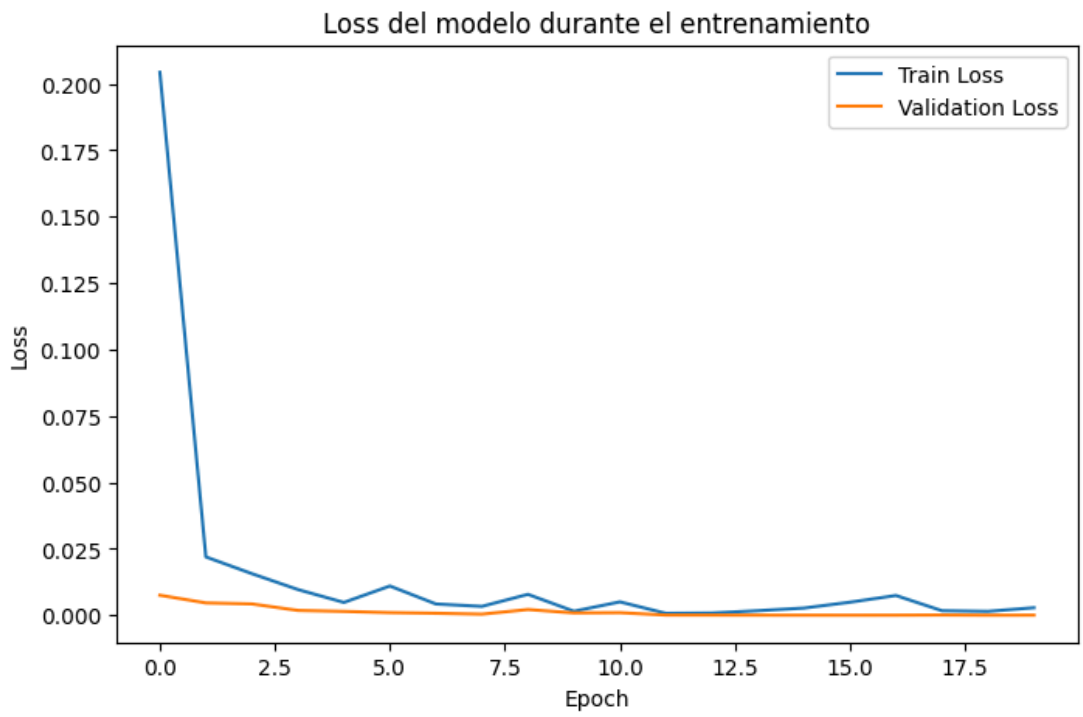


Gráfico de loss del entrenamiento

```
plt.figure(figsize=(8,5))
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Loss del modelo durante el entrenamiento')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```



7. Evaluación del modelo

```
loss, accuracy_dl = model.evaluate(X_test, y_test, verbose=0)
print('Accuracy Deep Learning:', accuracy_dl)
print('Loss Deep Learning:', loss)
```

```
Accuracy Deep Learning: 1.0
Loss Deep Learning: 5.344670353224501e-05
```

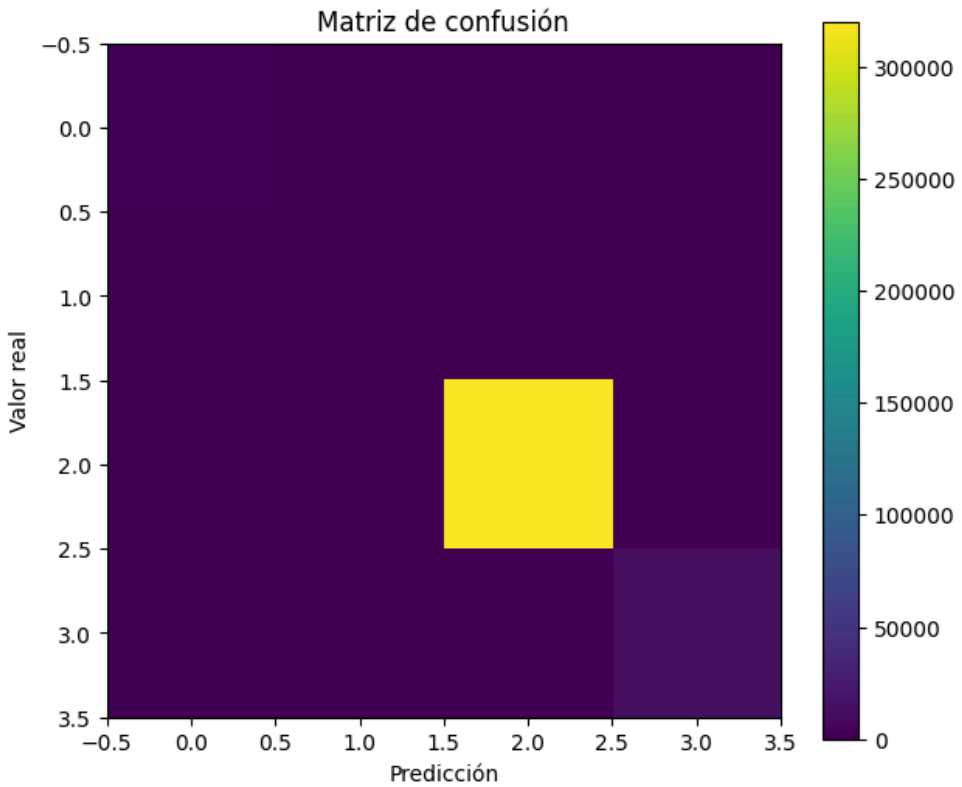
8. Resultados y métricas principales

```
y_pred_prob = model.predict(X_test, verbose=0)
y_pred_dl = y_pred_prob.argmax(axis=1)

print(classification_report(y_test, y_pred_dl))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      1753
           1       1.00      1.00      1.00       542
           2       1.00      1.00      1.00    319936
           3       1.00      1.00      1.00     14491

    accuracy                           1.00    336722
   macro avg       1.00      1.00      1.00    336722
weighted avg       1.00      1.00      1.00    336722
```

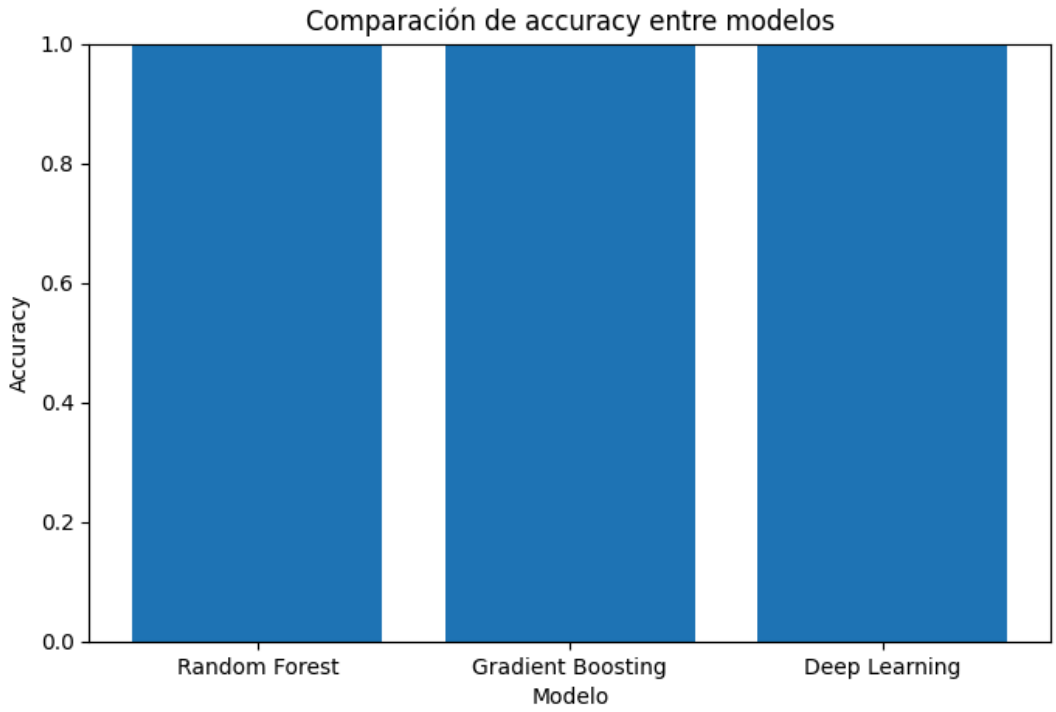
Matriz de confusión

```
cm = confusion_matrix(y_test, y_pred_dl)

plt.figure(figsize=(7,6))
plt.imshow(cm)
plt.title('Matriz de confusión')
plt.xlabel('Predicción')
plt.ylabel('Valor real')
plt.colorbar()
plt.show()
```



Comparación visual de accuracy entre modelos

```
modelos = ['Random Forest', 'Gradient Boosting', 'Deep Learning']
accuracies = [accuracy_rf, accuracy_gb, accuracy_dl]

plt.figure(figsize=(8,5))
plt.bar(modelos, accuracies)
plt.title('Comparación de accuracy entre modelos')
plt.xlabel('Modelo')
plt.ylabel('Accuracy')
plt.ylim(0, 1)
plt.show()
```



Interpretación de métricas

- **Accuracy:** porcentaje total de empresas clasificadas correctamente.
- **Precision:** exactitud del modelo al predecir cada categoría.
- **Recall:** capacidad del modelo para identificar correctamente cada tipo de empresa.
- **F1 Score:** balance entre precision y recall.
- **Matriz de confusión:** permite visualizar en qué categorías el modelo se equivoca más.